13.6 Errors That Add Up

WORKSHEET

The question

STEP II 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q15

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If a number's fractional part U is uniform on $[0, 1)$, what is the error when it is truncated? When it is rounded? What are the mean and variance of each?

STAGE 2 The probability of a large error

Ask yourself For LEP, what is $\mathrm{P}(|E| > 15)$ with $E \sim \mathrm{N}(0, 125)$? For VOZ, where is the error almost certainly?

STAGE 3 How many additions?

Ask yourself After n additions, LEP's error has standard deviation $\sqrt{n/12}$. When is $\mathrm{P}(|E| < 10) \geq 0.9$?

What's really going on

Bias and variance

Truncation is *biased*: its errors have non-zero mean, and bias accumulates linearly. Rounding is *unbiased*, and its errors behave like a random walk, growing like $\sqrt{n}$. After a million operations, the difference is between an error of order 500 000 and one of order 300. This is the same distinction as between a systematic error in an experiment (a miscalibrated instrument) and random measurement error: averaging many measurements reduces random error but does nothing to systematic error.

Floating point

Every computer stores real numbers to limited precision, and every arithmetic operation rounds its result. Modern computers follow the IEEE 754 standard, which rounds to the nearest representable number, with ties broken towards an even last digit precisely so that rounding is unbiased on average. Even so, careless summation of many numbers can lose accuracy; William Kahan's *compensated summation* algorithm (1965) keeps track of the lost low-order digits and adds them back. In the Gulf War in 1991, a Patriot missile battery failed to intercept an incoming missile because its clock counted time in tenths of a second, a number that cannot be stored exactly in binary; after 100 hours of operation the truncation error had grown to about a third of a second, enough to misjudge the target's position by over half a kilometre.

The Vancouver index

The Vancouver Stock Exchange index was created in January 1982 with a value of 1000.000. By November 1983 it had fallen to about 524, despite a generally rising market. Consultants found that the index was being recalculated thousands of times a day and truncated (not rounded) to three decimal places each time. Recomputed correctly over a weekend, the index reopened at about 1098. The episode is a standard warning in courses on numerical computation.

Try this next

1. **VOZ corrected.** If VOZ adds $\frac{n}{2}$ to its answer at the end, what is the distribution of its error? Is it now as good as LEP?

2. **Rounding to one decimal place.** If numbers are rounded to one decimal place, with uniformly distributed discarded digits, find the standard deviation of the total error in adding 10 000 numbers.

3. **The Vancouver index.** If the index was recalculated 3000 times a day and each truncation lost on average 0.0005, how much would it lose in a year of 250 trading days?

4 **The exact distribution.** For the sum of two rounding errors, find the exact density (a triangle). For three, sketch it. How quickly does it start to look normal?

Further reading

- **Nicholas Higham, *Accuracy and Stability of Numerical Algorithms*** (SIAM). The authoritative book on rounding errors; the opening chapter is accessible.
- **David Goldberg, "What every computer scientist should know about floating-point arithmetic"**, *ACM Computing Surveys* 23 (1991). A classic, free online.
- **Tim Harford, *How to Make the World Add Up*** (Bridge Street), on systematic errors in data.

The full guided solution is in Section 13.6:
https://beyondstep.keithmcnulty.org/ch13/errors-add-up.html